

Article

Numerical Simulation of Preheating Temperature on Molten Pool Dynamics in Laser Deep-Penetration Welding

Jin Peng ^{1,*} , Jigao Liu ¹, Xiaohong Yang ^{2,3,*}, Jianya Ge ^{2,3}, Peng Han ¹, Xingxing Wang ¹, Shuai Li ¹  and Yongbiao Wang ⁴

¹ International Joint Laboratory of High-Efficiency Special Green Welding, Material College, North China University of Water Resources and Electric Power, Zhengzhou 450045, China

² Jinhua Polytechnic, Jinhua 321017, China

³ Key Laboratory of Crop Harvesting Equipment Technology of Zhejiang Province, Jinhua 321017, China

⁴ Henan Key Laboratory of Mechanical Equipment Intelligent Manufacturing, Zhengzhou Light Industry University, Zhengzhou 450002, China

* Correspondence: pengjin1985120@163.com (J.P.); 20050626@jhc.edu.cn (X.Y.)

Abstract: In this paper, a heat-flow coupling model of laser welding at preheating temperature was established by the FLUENT 19.0 software. The fluctuation of the keyhole wall and melt flow behavior in the molten pool under different preheating temperatures were analyzed, and the correlation between keyhole wall fluctuation and molten pool flow with spatters and bubbles was obtained. The results indicate that when the outer wall in the middle of the rear keyhole wall is convex, the inner wall is concave, which causes spatter or the bottom of the keyhole to collapse. When the metal layer in the middle of the rear keyhole wall turns into obliquely upward flow, welding spatter is generated. In contrast, the metal layer in the middle of the rear keyhole wall changes to flow into the keyhole, and the bottom of the keyhole collapses. When the preheating temperature is 300 K (ambient temperature), 400 K, and 500 K, the inner wall in the middle of the rear keyhole wall is concave. With the increase in the preheating temperature, the area of the concave gradually increases, and the size of the liquid column behind the keyhole opening gradually decreases. When the preheating temperature is 300 K, there are more spatters above the molten pool. In comparison, when the preheating temperature is 400 K or 500 K, there are less spatters, and the bottom of the keyhole collapses.

Keywords: laser welding; numerical simulation; preheating temperature; molten pool



Citation: Peng, J.; Liu, J.; Yang, X.; Ge, J.; Han, P.; Wang, X.; Li, S.; Wang, Y. Numerical Simulation of Preheating Temperature on Molten Pool Dynamics in Laser Deep-Penetration Welding. *Coatings* **2022**, *12*, 1280. <https://doi.org/10.3390/coatings12091280>

Academic Editor: Stefanos M. Skolianos

Received: 4 July 2022

Accepted: 30 August 2022

Published: 1 September 2022

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1. Introduction

Compared with traditional welding methods, laser welding has the advantages of controllable energy accuracy, a small heat-affected zone, and low residual stresses [1–3]. Laser welding is a complex physical, chemical, and metallurgical process [4,5]. The dynamic fluctuation of the keyhole and flow behavior of liquid metal in the molten pool have a great influence on weld quality [6,7].

Through the establishment of a high-speed X-ray transmission camera system, Matsunawa et al. [8,9] studied the melt flow behavior of the molten pool in the laser welding of aluminum alloys. They found that the keyhole is always in a fluctuating state after it was formed, which causes the generation of bubbles. Pang et al. [10] established a three-dimensional model of double-beam laser welding a 5052 aluminum alloy. They found that along the welding direction, the keyhole depth formed by the latter laser beam is greater than that of the lead one, due to the preheating effect. Zhang et al. [11] developed a three-dimensional numerical model to investigate the flow behavior of molten pool and spatter formation in 5083 aluminum alloys. They found that welding spatters usually form at the keyhole opening, and gradually separate from the weld pool.

Han et al. [12] found that surface tension, buoyancy, and gravity affect the molten pool, but recoil pressure is the main driving force for the formation of keyholes, has a great

influence on the melt flow around the keyhole, and also affects the opening size of the keyhole. Geiger et al. [13] established a three-dimensional model of molten pool flow in laser beam welding. They found that the liquid metal in front of the keyhole wall flows downward from the surface of the molten pool, which causes the periodic fluctuation of the keyhole diameter and the collapse of the keyhole. Sohail et al. [14] found that flow vortices are created in the molten pool during laser welding, and the flow vortex becomes larger when the laser power increases. This was mainly because after increasing the heat input, the bottom of the keyhole absorbs more energy and accelerates the melt flow at the bottom of the keyhole.

As aluminum alloys have a high reflectivity to laser beam, preheating before welding could obviously increase the absorption of laser energy by the aluminum alloy, improves the utilization rate of laser energy, adjust the stress distribution of the welded joints, and improve the mechanical properties of welded joints. However, there are few studies on the correlation between keyhole wall fluctuation and molten pool flow with welding spatters and bubbles under different preheating temperatures.

In this paper, a heat-flow coupling model of laser welding at preheating temperature was established by the FLUENT 19.0 software. The fluctuation of the keyhole wall and melt flow behavior in the molten pool under different preheating temperatures were analyzed, and the correlation between keyhole wall fluctuation and molten pool flow with spatters and bubbles was obtained.

2. Mathematical Modeling

Partial penetrations were investigated in bead-on-plate welding of a 6056 aluminum alloy. In this mathematical model, the laser power was 2800 W, and the welding speed was 3 m/min. Before the welding, the temperature of the workpiece was 300 K (ambient temperature), 400 K, and 500 K. The influence of welding driving forces was considered in the mathematical model. The main driving forces of the weld pool were obtained from the literature [15].

2.1. Numerical Model and Computational Assumptions

The calculation region of the laser welding process was established, as shown in Figure 1. Due to the symmetry of the weld, the calculation area was only half of the test piece, in order to improve the calculation efficiency. The length direction of the test piece was the x-axis, the width direction was the positive direction of the y-axis, and the thickness direction was the negative direction of the z-axis. The calculation area consisted of two parts, the upper part was the air area, and the lower part was the metal area.

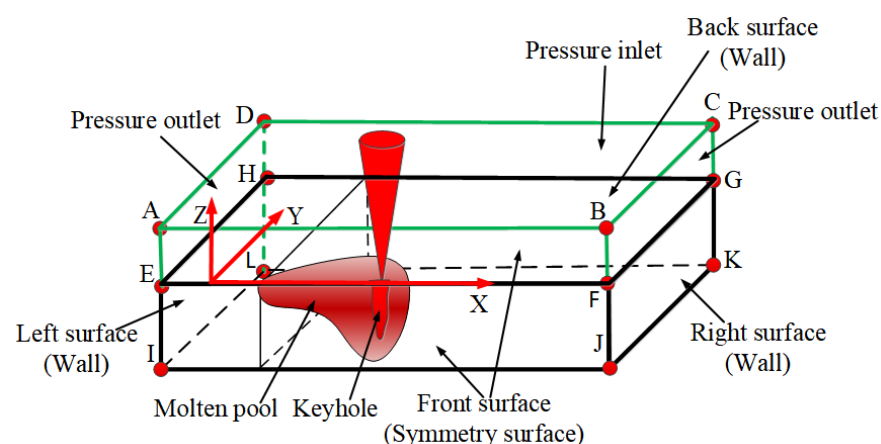


Figure 1. The calculation region of the laser welding.

2.2. Heat Source Model

In this paper, the laser beam was treated as a rotating Gaussian body heat source. The laser welding heat source model expression is shown by Equation (1) [16]:

$$q_{laser} = \frac{9\alpha_{abs}Q}{\pi R_0^2 H(1-e^3)} \exp\left[\frac{-9(x^2 + y^2)}{R_0^2 \log(H/z)}\right] \quad (1)$$

where α_{abs} is the percentage of incident laser energy absorbed by the keyhole, Q is the energy of the laser heat source, H is the height of the laser heat source; R_0 is the effective radius of the laser beam, and x , y , and z denote the coordinates of the laser beam during laser welding.

2.3. Governing Equations

The continuity equation is shown:

$$\frac{\partial(\rho)}{\partial t} + \frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} + S_m = 0 \quad (2)$$

The energy equation is shown:

$$\begin{aligned} \frac{\partial(\rho H)}{\partial t} + \frac{\partial(\rho u H)}{\partial x} + \frac{\partial(\rho v H)}{\partial y} + \frac{\partial(\rho w H)}{\partial z} &= \frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(k \frac{\partial T}{\partial y} \right) \\ &+ \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right) + S_H \end{aligned} \quad (3)$$

The momentum conservation equation is expressed as follows:

$$\begin{aligned} \frac{\partial(\rho u)}{\partial t} + \frac{\partial(\rho u u)}{\partial x} + \frac{\partial(\rho u v)}{\partial y} + \frac{\partial(\rho u w)}{\partial z} &= \frac{\partial}{\partial x} \left(\mu \frac{\partial u}{\partial x} \right) + \frac{\partial}{\partial y} \left(\mu \frac{\partial u}{\partial y} \right) \\ &+ \frac{\partial}{\partial z} \left(\mu \frac{\partial u}{\partial z} \right) - \frac{\partial P}{\partial x} + S_x \end{aligned} \quad (4)$$

$$\begin{aligned} \frac{\partial(\rho v)}{\partial t} + \frac{\partial(\rho u v)}{\partial x} + \frac{\partial(\rho v v)}{\partial y} + \frac{\partial(\rho v w)}{\partial z} &= \frac{\partial}{\partial x} \left(\mu \frac{\partial v}{\partial x} \right) + \frac{\partial}{\partial y} \left(\mu \frac{\partial v}{\partial y} \right) \\ &+ \frac{\partial}{\partial z} \left(\mu \frac{\partial v}{\partial z} \right) - \frac{\partial P}{\partial y} + S_y \end{aligned} \quad (5)$$

$$\begin{aligned} \frac{\partial(\rho w)}{\partial t} + \frac{\partial(\rho u w)}{\partial x} + \frac{\partial(\rho v w)}{\partial y} + \frac{\partial(\rho w w)}{\partial z} &= \frac{\partial}{\partial x} \left(\mu \frac{\partial w}{\partial x} \right) + \frac{\partial}{\partial y} \left(\mu \frac{\partial w}{\partial y} \right) \\ &+ \frac{\partial}{\partial z} \left(\mu \frac{\partial w}{\partial z} \right) - \frac{\partial P}{\partial z} + S_z \end{aligned} \quad (6)$$

where u , v , and w are the velocity vectors of different coordinate axes directions in the mathematical model; ρ is the density, P is the pressure, H is the enthalpy, k is the thermal conductivity, and μ is the viscosity; S_m , S_x , S_y , S_z , and S_H are the source terms of the governing equations.

In this mathematical model, the welding base material was 6056 aluminum alloy, and its thermophysical parameters were obtained from the literature [17], as shown in Table 1.

$$C_p \left(\text{Jkg}^{-1}\text{K}^{-1} \right) = \begin{cases} -0.001 \times T^2 + 1.1609 \times T + 267.71 & 300 < T \leq 573 \\ 0.0009 \times T^2 - 0.3901 \times T + 514.45 & 573 < T \leq 913 \\ -0.0009 \times T^2 + 0.5832 \times T + 435.14 & 913 < T \leq 2740 \end{cases} \quad (7)$$

$$k \left(\text{Wm}^{-1}\text{K}^{-1} \right) = \begin{cases} -0.0001 \times T^2 - 0.0697 \times T + 95.334 & 300 < T \leq 860 \\ -0.0048 \times T^2 + 9.2812 \times T - 4275.6 & 860 < T \leq 917 \\ -0.00001 \times T^2 + 0.0582 \times T + 148.74 & 917 < T \leq 2740 \end{cases} \quad (8)$$

$$\mu \left(\text{kgm}^{-1}\text{K}^{-1} \right) = \begin{cases} 1 \times 10^{-7} \times T^2 - 0.0002 \times T + 0.1202 & 897 < T \leq 937 \\ 2 \times 10^{-11} \times T^2 - 5 \times 10^{-7} \times T + 0.0038 & 937 < T \leq 2650 \\ -6 \times 10^{-8} \times T^2 + 0.0003 \times T - 0.4151 & 2650 < T \leq 2720 \end{cases} \quad (9)$$

Table 1. Physical and thermal properties of 6056 aluminum alloy.

Property	Symbol	Unit	Value
Solid density	ρ_s	$\text{kg}\cdot\text{m}^{-3}$	2720
Liquid density	ρ_l	$\text{kg}\cdot\text{m}^{-3}$	2590
Solidus temperature	T_s	K	860
Liquidus temperature	T_L	K	917
Boiling temperature	T_g	K	2740
Latent heat of fusion	L_m	$\text{J}\cdot\text{kg}^{-1}$	3.87×10^5
Latent heat of the vapor	L_v	$\text{J}\cdot\text{kg}^{-1}$	1.08×10^7
Thermal expansion coefficient	β_k	K^{-1}	1.92×10^{-5}
Convective heat transfer coefficient	h_0	$\text{W}\cdot\text{K}^{-1}\cdot\text{m}^{-2}$	15
Surface tension	δ_0	$\text{N}\cdot\text{m}^{-1}$	0.914
Surface tension gradient	A_δ	$\text{N}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$	-3.5×10^{-4}
Radiation emissivity	ε	-	0.08
Ambient temperature	T_{ref}	K	300

3. Results and Discussion

3.1. Comparison between the Calculated and the Experimental Result

Figure 2 shows the comparison of the weld cross-section obtained by experiment and the molten pool obtained by numerical simulation when the preheating temperature is 300 K (ambient temperature). It is found that the shape and dimensions of the numerically obtained weld pool are in good agreement with those obtained experimentally.

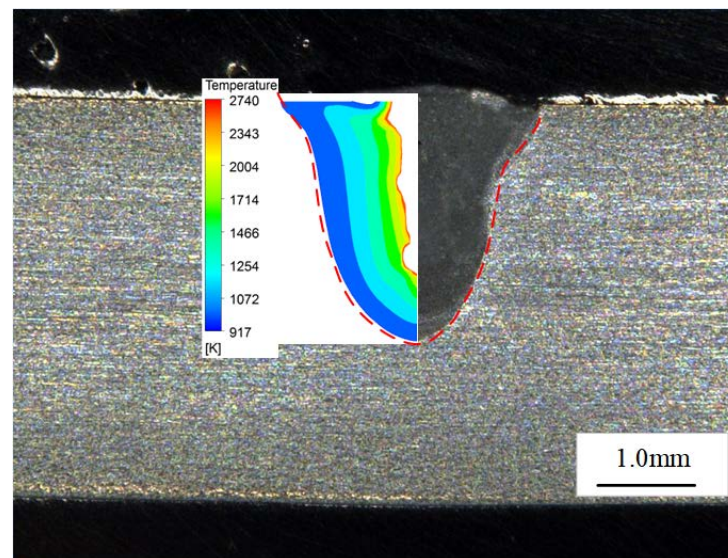


Figure 2. Comparison of the weld seam shape (right) and the simulated molten pool (left) for laser welding with preheating temperature of 300 K (ambient temperature).

3.2. The Dynamic Fluctuation of Keyhole Wall and the Melt Flow on Keyhole Wall

To display the flow field of the keyhole wall, the three-dimensional keyhole was cut along the XOZ plane. Figure 3 is the schematic of the intercepting 3D keyhole, and the yellow area is the section of intercepting 3D keyhole.

In this paper, the dynamic fluctuation of the keyhole wall and the melt flow on the keyhole wall were studied and analyzed. Figures 4–6 show the dynamic fluctuation of the keyhole wall and the melt flow on the keyhole wall with preheating temperatures of 300 K (ambient temperature), 400 K, and 500 K, respectively.

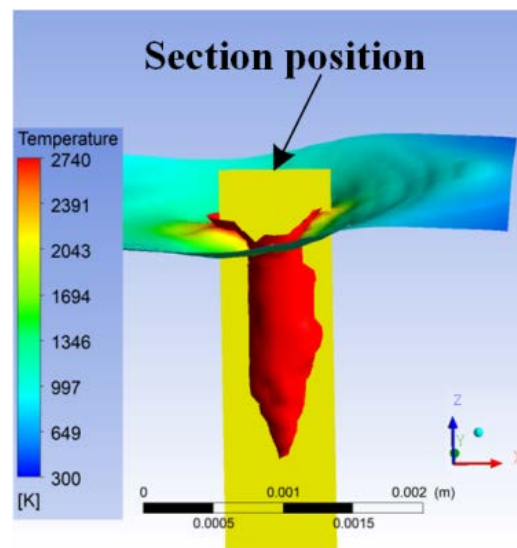


Figure 3. Section position of the keyhole wall.

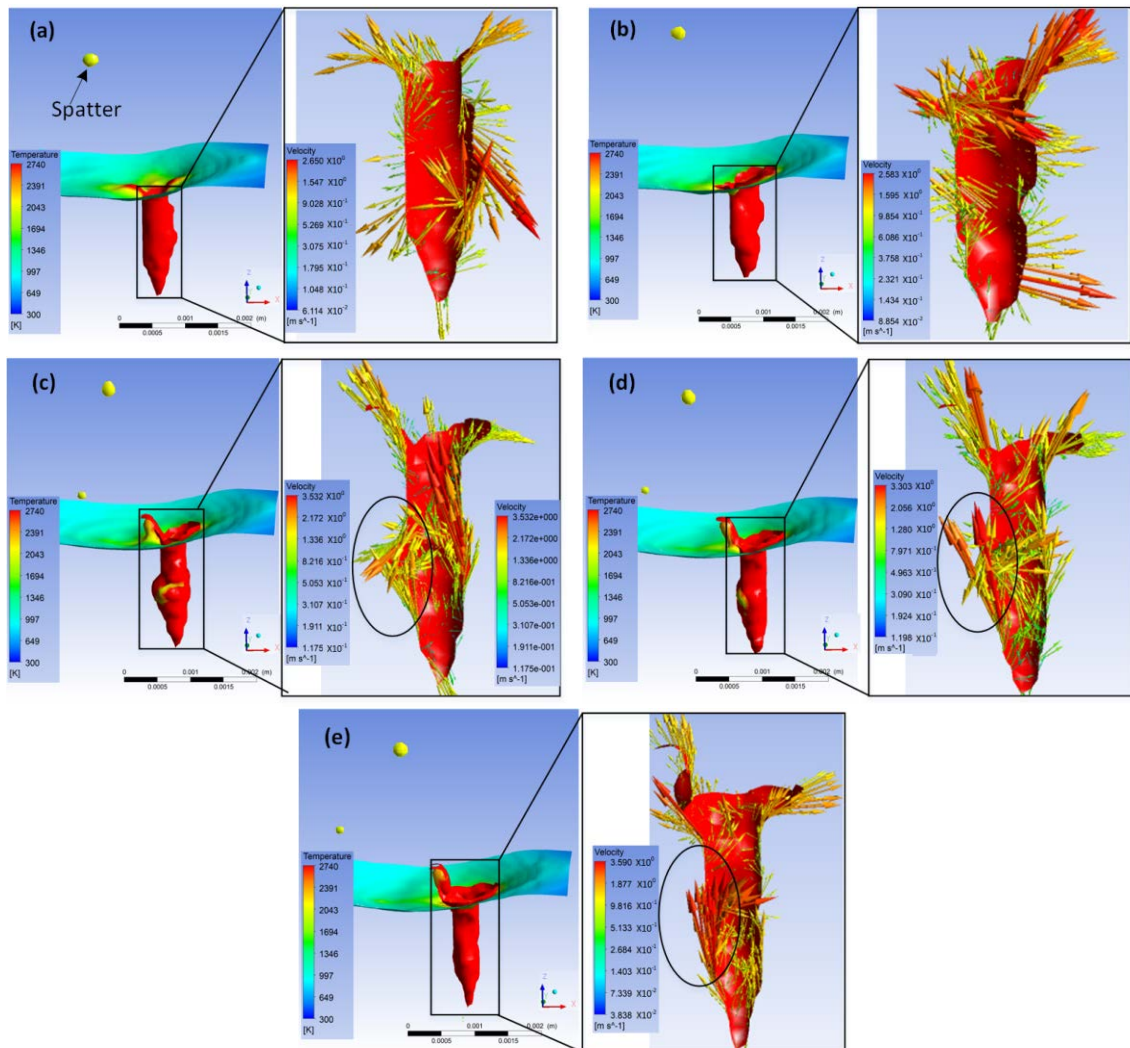


Figure 4. The dynamic fluctuation of keyhole wall and the melt flow on keyhole wall with pre-heating temperature of 300 K (ambient temperature): (a) $t = 26.4$ ms, (b) $t = 28$ ms, (c) $t = 29$ ms, (d) $t = 29.1$ ms, (e) $t = 29.2$ ms.

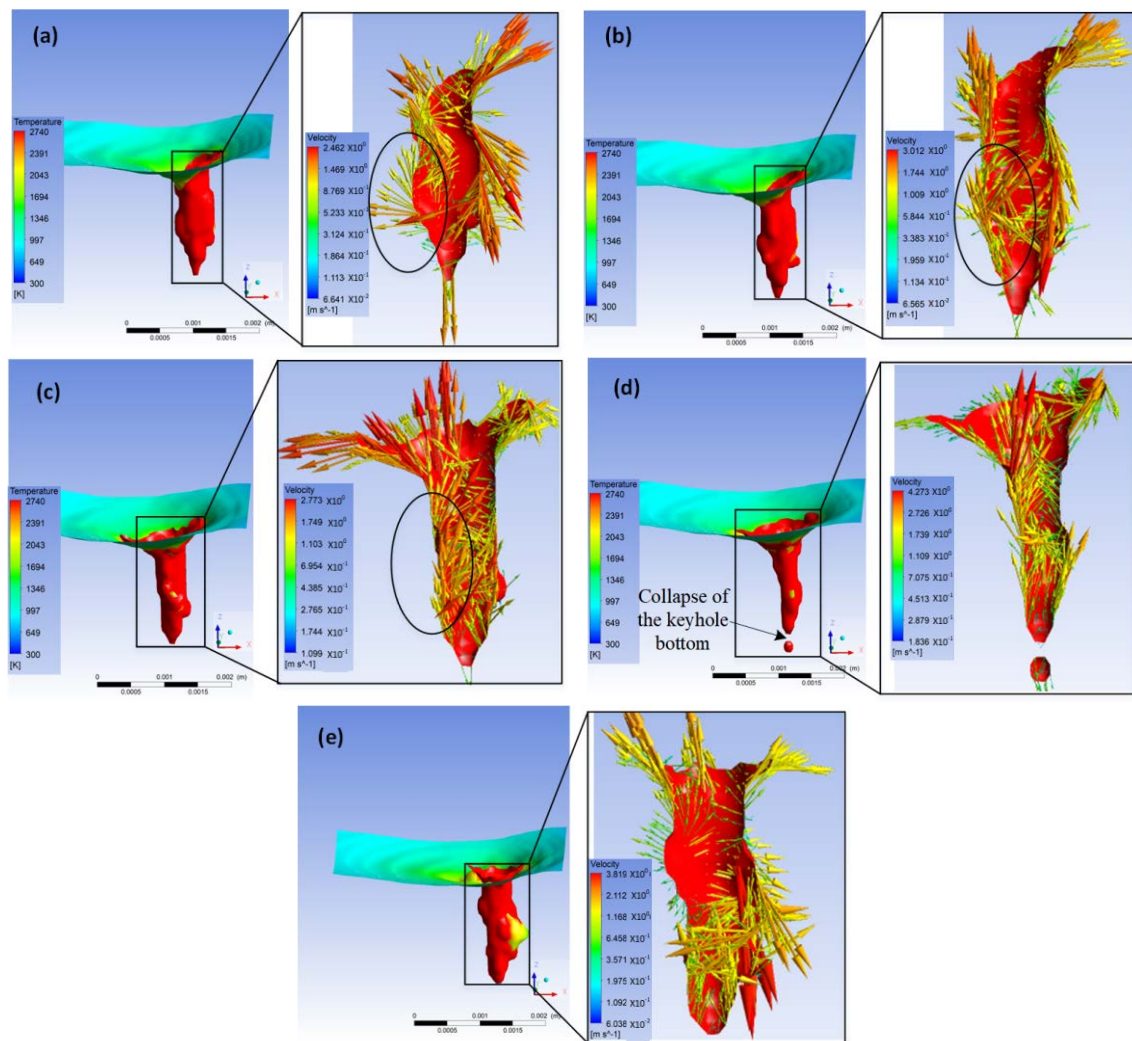


Figure 5. The dynamic fluctuation of keyhole wall and the melt flow on keyhole wall with preheating temperature of 400 K: (a) $t = 26.4$ ms, (b) $t = 26.5$ ms, (c) $t = 26.7$ ms, (d) $t = 26.8$ ms, (e) $t = 29$ ms.

Through comparison, it is found that the size in the middle of the keyhole gradually increases with the increase in the preheating temperature. When the preheating temperature is 300 K (ambient temperature), the maximum size in the middle part of the keyhole is 0.57 mm, and the minimum size is 0.37 mm. When the preheating temperature is 400 K, the maximum size in the middle part of the keyhole is 0.6 mm, and the minimum size is 0.33 mm. When the preheating temperature is 500 K, the maximum size in the middle part of the keyhole is 0.68 mm, and the minimum size is 0.29 mm. The fluctuation of the keyhole size affects the stability of the welding process, and causes welding defects [18–21].

When the outer wall in the middle of the rear keyhole wall is convex, the inner wall is concave, which causes spatter or the bottom of the keyhole to collapse. When the preheating temperature increases from 300 K to 500 K, the metal layer in the middle of the rear keyhole wall flows outward (the oval area in Figures 4c, 5a and 6a). This flow trend is beneficial in maintaining the stability of the keyhole [22–24]. However, with the progress of the welding process, when the metal layer in the middle of the rear keyhole wall turns into obliquely upward flow (Figure 4e), welding spatter is generated. In contrast, the metal layer in the middle of the rear keyhole wall changes to flow into the keyhole (Figures 5c and 6c), and the bottom of the keyhole collapses (Figures 5d and 6d).

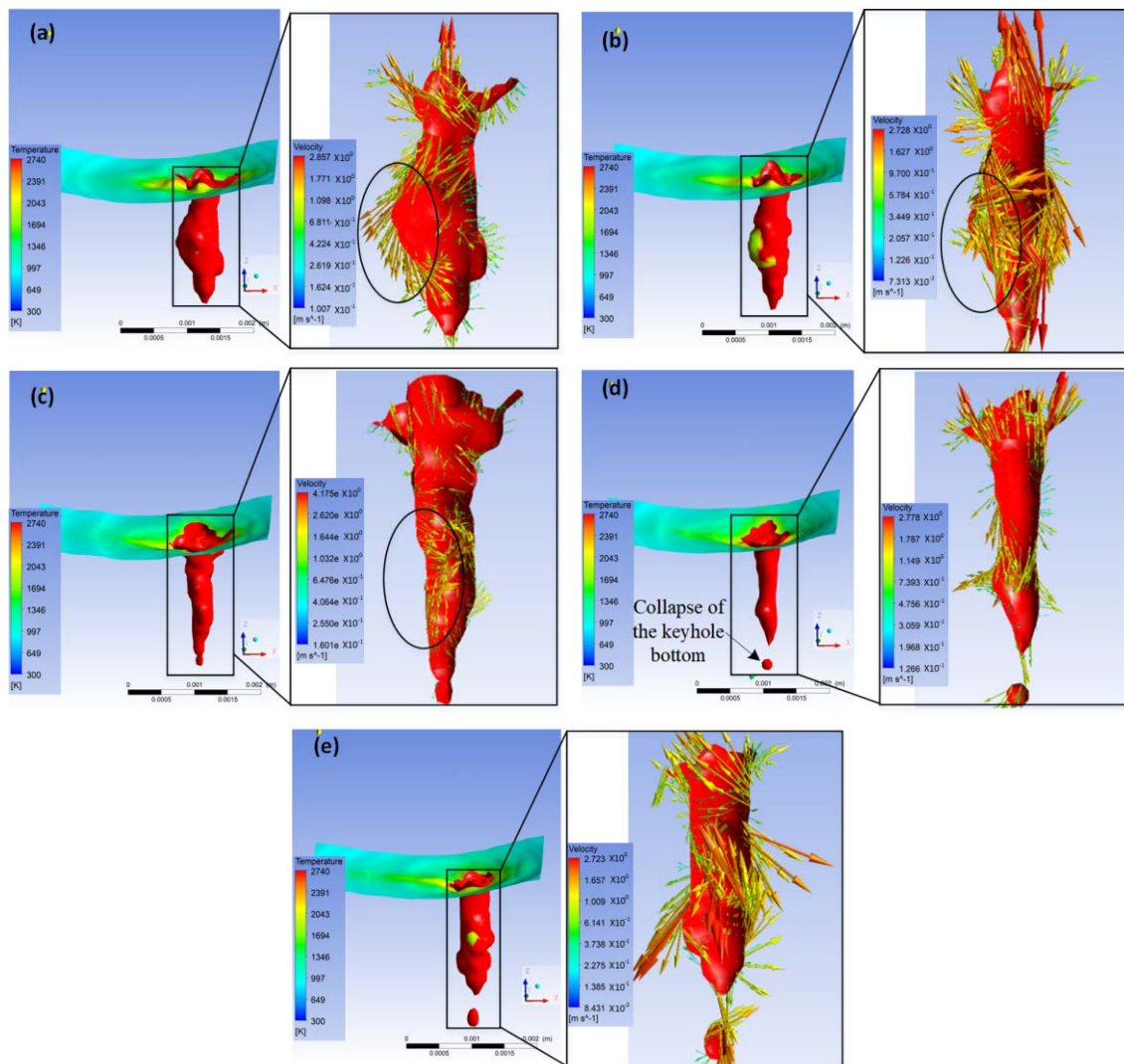


Figure 6. The dynamic fluctuation of keyhole wall and the melt flow on keyhole wall with preheating temperature of 500 K: (a) $t = 28$ ms, (b) $t = 28.2$ ms, (c) $t = 28.5$ ms, (d) $t = 28.7$ ms, (e) $t = 29$ ms.

3.3. The Melt Flow Behavior of the Molten Pool

The melt flow behavior in the molten pool with different preheating temperatures was further studied in this paper. Figures 7–9 show the melt flow behavior in the molten pool with preheating temperatures of 300 K (ambient temperature), 400 K, and 500 K, separately.

As shown in Figure 7, when the preheating temperature is 300 K (ambient temperature), there are more spatters above the molten pool (Figure 7c,e). In comparison, when the preheating temperature is 400 K or 500 K (Figures 8 and 9, respectively), there are less spatters.

When the welding time is 29 ms (Figures 7c, 8e and 9e), as the preheating temperature increases, the size of the liquid column behind the keyhole opening gradually decreases. Moreover, it is seen from Figure 10 that with the increase in the preheating temperature, the flow velocity of the liquid column behind the keyhole opening gradually decreases, which indicates that the tendency to form spatter becomes weaker.

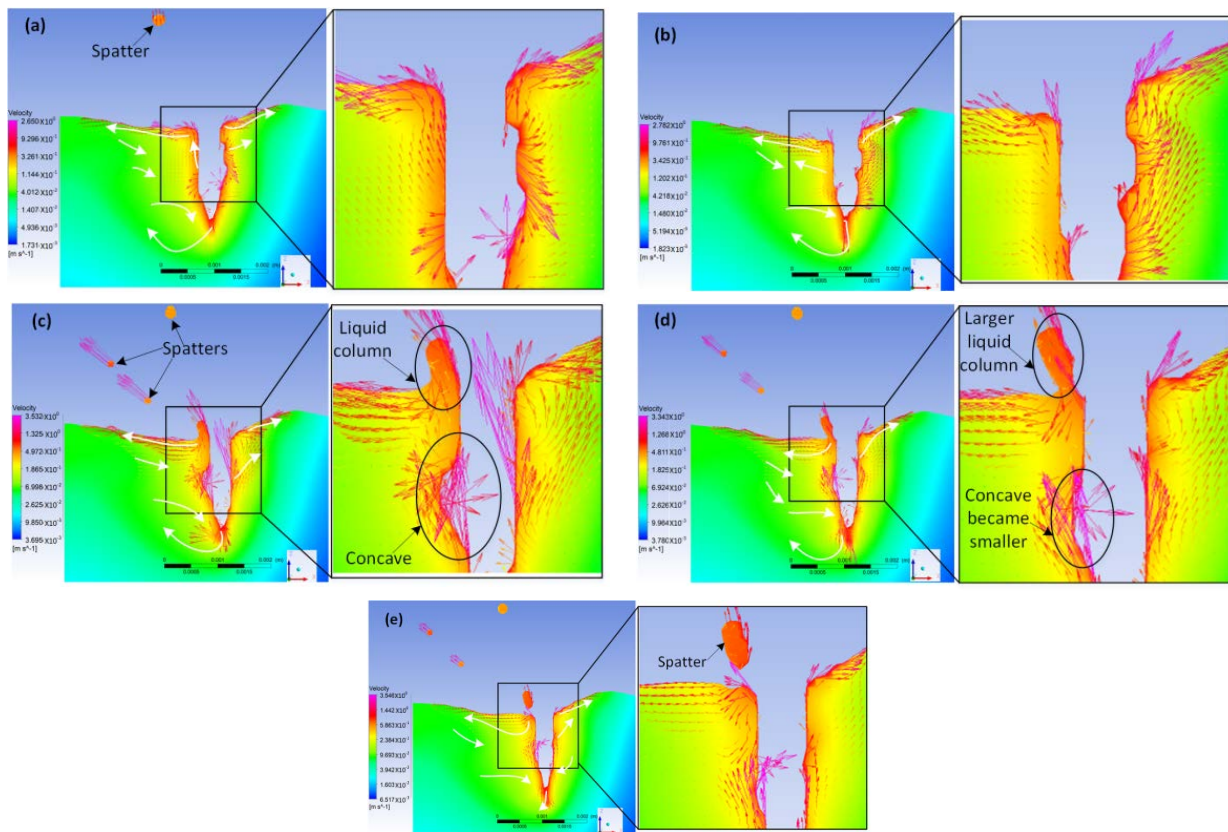


Figure 7. The melt flow behavior in the molten pool with preheating temperature of 300 K (ambient temperature): (a) $t = 26.4$ ms, (b) $t = 28$ ms, (c) $t = 29$ ms, (d) $t = 29.1$ ms, (e) $t = 29.2$ ms.

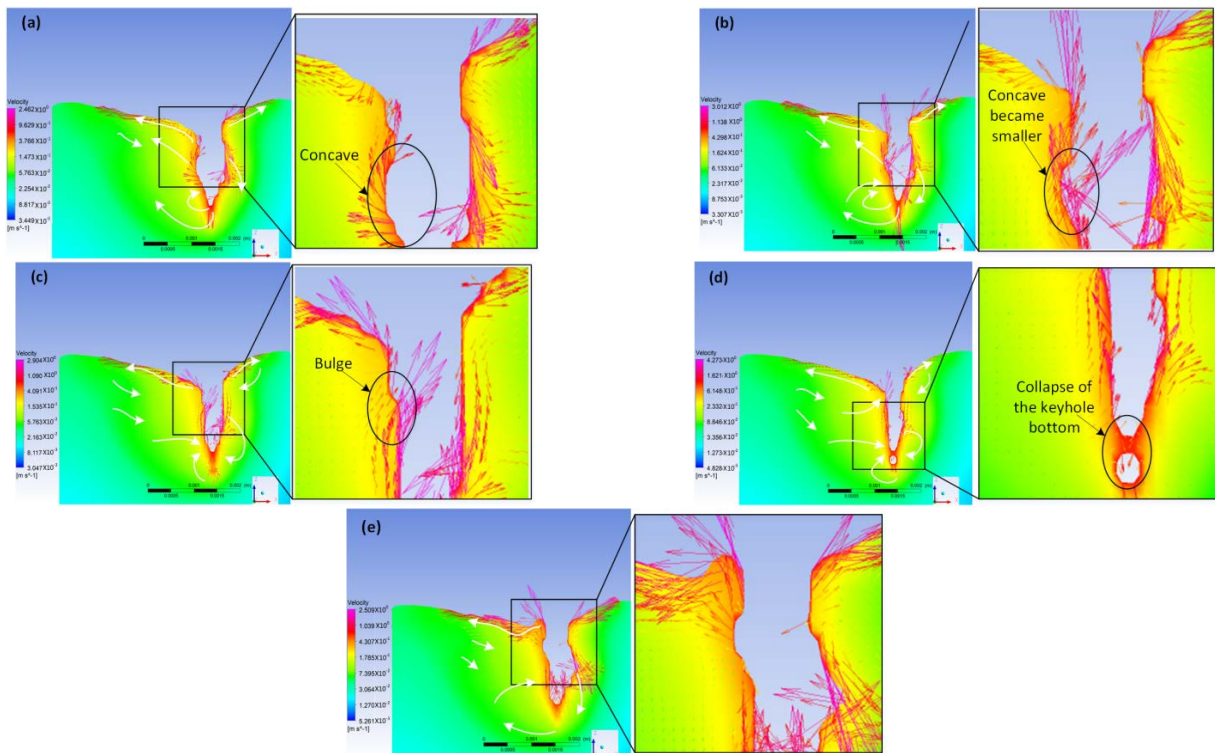


Figure 8. The melt flow behavior in the molten pool with preheating temperature of 400 K: (a) $t = 26.4$ ms, (b) $t = 26.5$ ms, (c) $t = 26.7$ ms, (e) $t = 26.8$ ms, (e) $t = 29$ ms.

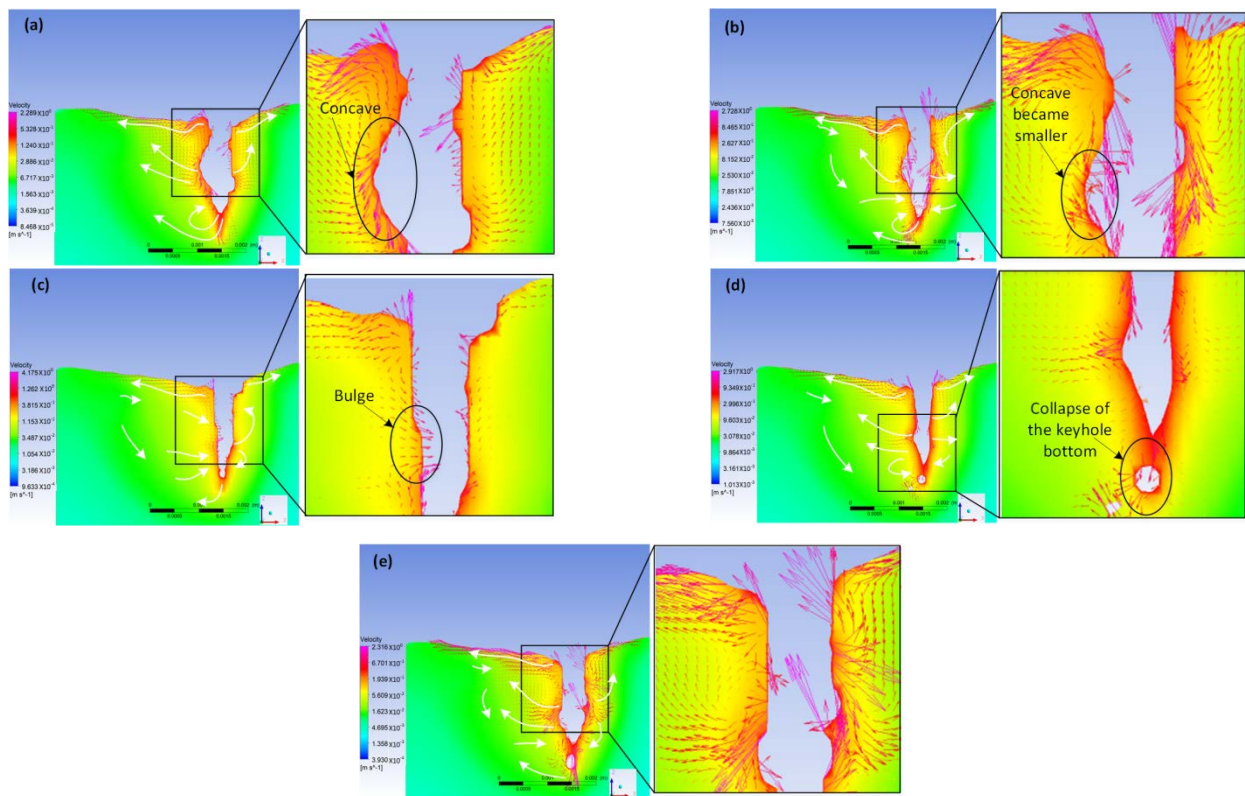


Figure 9. The melt flow behavior in the molten pool with preheating temperature of 500 K: (a) $t = 28$ ms, (b) $t = 28.2$ ms, (c) $t = 28.5$ ms, (d) $t = 28.7$ ms, (e) $t = 29$ ms.

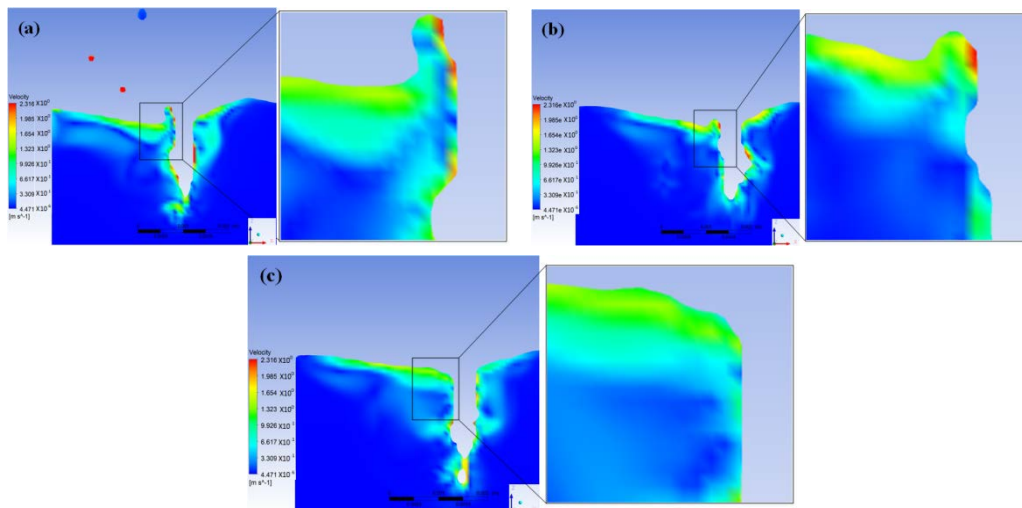


Figure 10. Flow velocity nephogram of molten pool when the welding time is 29 ms: (a) the preheating temperature is 300 K (ambient temperature), (b) the preheating temperature is 400 K, (c) the preheating temperature is 500 K.

When the preheating temperature is 300 K (ambient temperature), 400 K, and 500 K, the inner wall in the middle of the rear keyhole wall is concave (Figures 7c, 8a and 9a), and with the increase in the preheating temperature, the area of the concave gradually increases.

When the preheating temperature is 300 K, with the progress of welding, the concave area in the middle of the rear keyhole wall decreases gradually (Figure 7c,d), and the liquid column behind the keyhole opening increases gradually, and finally forms the spatter

(Figure 7e). When the preheating temperature is 400 K and 500 K, with the progress of welding, the concave area in the middle of the rear keyhole wall decreases gradually (Figures 8a,b and 9a,b), and then turns to bulge (Figures 8c and 9c), and collapses at the bottom of the keyhole (Figures 8d and 9d).

Figures 11 and 12 show the process of generating spatter and collapsing at the bottom of the keyhole, respectively. As shown in Figure 11, under the action of recoil pressure, the concave area appears in the middle of the inner wall of the keyhole. With the progress of welding, the concave area in the middle of the rear keyhole wall decreases gradually, and the liquid column behind the keyhole opening increases gradually, and finally forms the spatter. As shown in Figure 12, under the action of recoil pressure, the concave area appears in the middle of the inner wall of the keyhole. With the progress of welding, the concave area in the middle of the rear keyhole wall decreases gradually, and then turns to bulge, and collapses at the bottom of the keyhole. Therefore, bubbles form in the molten pool.

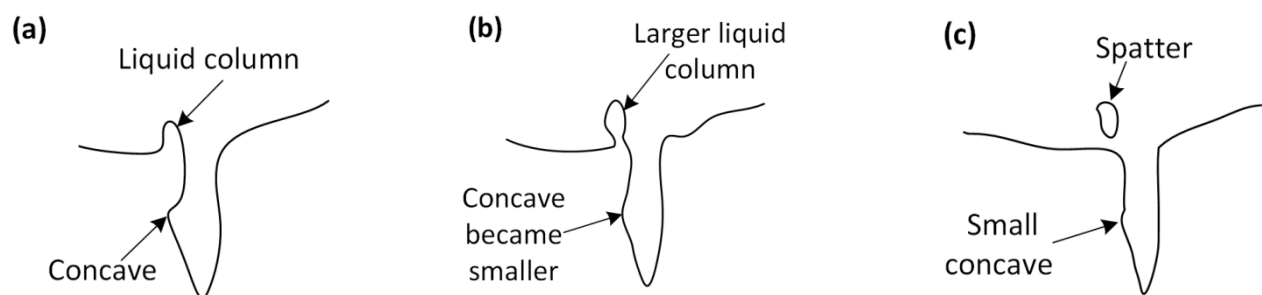


Figure 11. The process of generating spatter around the keyhole: (a) the concave area forms in the middle of the inner wall of the keyhole, (b) the concave area becomes smaller in the middle of the keyhole wall, (c) the spatter forms at the opening of keyhole.

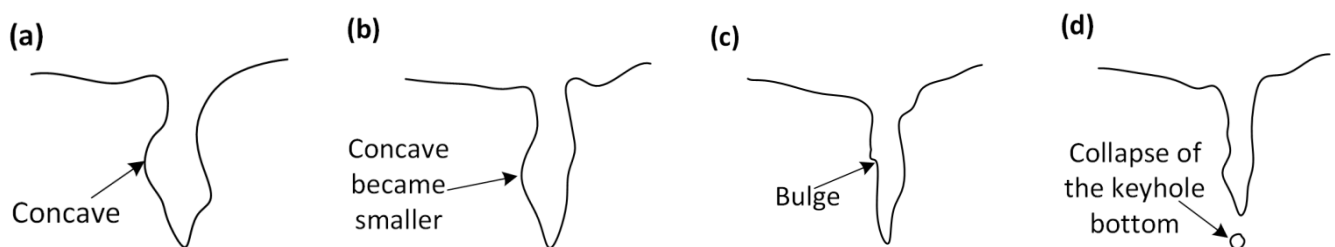


Figure 12. The process of collapsing at the bottom of the keyhole: (a) the concave area forms in the middle of the inner wall of the keyhole, (b) the concave area becomes smaller in the middle of the keyhole wall, (c) the bulge forms in the middle of the keyhole wall, (d) collapse of the keyhole bottom and formation of the bubble.

When the laser beam acts on the base metal, the keyhole is formed with the vaporization of the base metal. When the length of the keyhole is greater than the circumference ($l > 2\pi r$, l : the keyhole depth, r : the keyhole radius) of the keyhole, the stability of the keyhole decreases. At this time, the keyhole is easy to collapse and close in the length direction, which is called Rayleigh instability. When the keyhole is closed, the laser energy acts on the closed area of the keyhole to evaporate the liquid metal in the area, and the keyhole opens under the action of the recoil pressure, as shown in Figure 13. In the process of laser welding, the shape of the keyhole is in a state of dynamic change. When the shape of keyhole fluctuates to a large extent, it is easy to cause welding defects (such as bubbles or spatters).

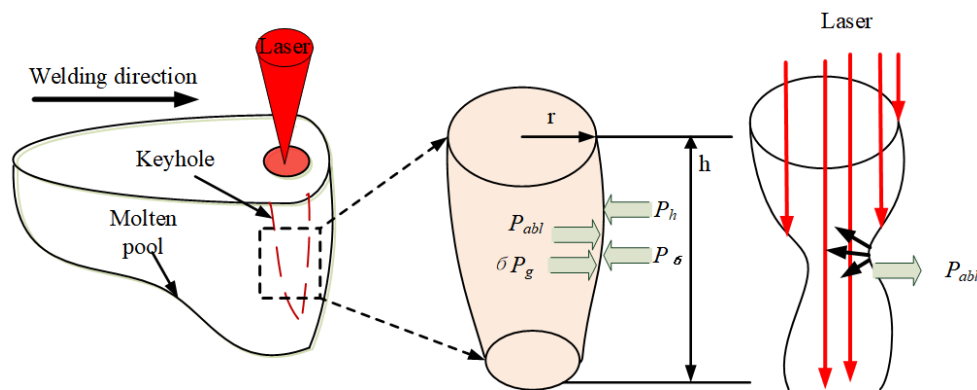


Figure 13. Relationship between the deformation of the keyhole wall and the acting force.

When the keyhole is in a stable state, the force acting on the keyhole wall is as follows:

$$P_{abl} + \delta P_g = P_\sigma + P_h \quad (10)$$

where P_{abl} , δP_g , P_σ , and P_h meant the recoil pressure, the excess vapor pressure, the surface tension pressure, and the hydrostatic pressure, respectively.

Due to preheating, the absorption rate of the laser increases and the energy required for melting the metal is reduced. Under the same welding parameters, due to the effect of preheating before welding, the energy acting on the keyhole increases, and the metal vaporization rate on the keyhole wall increases, so the recoil pressure increases. The force on the left side of Formula (10) is greater than the force on the right side of Formula (10). Therefore, there is a concave area in the inner wall of the keyhole and a convex area on the outer wall of the keyhole.

Figure 14 shows the schematic diagram of the molten pool along the welding direction during laser welding. The incident beam has a uniform intensity I_0 , with a diameter D . V_d is the ‘drilling velocity’, and V_w is the welding speed.

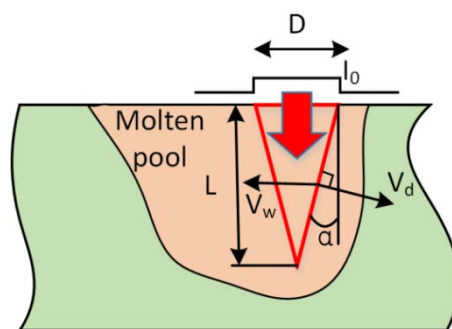


Figure 14. Schematic diagram of the molten pool along the welding direction during laser welding.

The simplified formula for the inclination angle α of the keyhole front wall is as follows:

$$\tan \alpha \approx V_w / (k I_0 A_0) \quad (11)$$

The formula for the penetration depth L during laser welding is as follows [15]:

$$L \approx k I_0 A_0 D / V_w = (4 A_0 k / \pi) P / (D V_w) \quad (12)$$

where P is the laser power, and A_0 is the absorptivity under normal incidence. When the preheating temperature increases, the absorptivity of the laser energy by the base metal increases, so the L increases. However, when the L increases, the stability of the keyhole decreases, and the bottom of the keyhole is prone to collapse.

4. Conclusions

1. When the outer wall in the middle of the rear keyhole wall is convex, the inner wall is concave, which causes spatter or the bottom of the keyhole to collapse. When the metal layer in the middle of the rear keyhole wall turns into obliquely upward flow, welding spatter is generated. In contrast, the metal layer in the middle of the rear keyhole wall changes to flow into the keyhole, and the bottom of the keyhole collapses;

2. When the preheating temperature is 300 K (ambient temperature), 400 K, or 500 K, the inner wall in the middle of the rear keyhole wall is concave. With the increase in the preheating temperature, the concave area gradually increases, and the size of the liquid column behind the keyhole opening gradually decreases;

3. When the preheating temperature is 300 K, there are more spatters above the molten pool. In comparison, when the preheating temperature is 400 K or 500 K, there are less spatters, and the bottom of the keyhole collapses.

Author Contributions: Conceptualization, J.P., J.L. and P.H.; Data curation, J.P.; Formal analysis, J.P.; Funding acquisition, J.P., X.Y., J.G. and X.W.; Investigation, J.P., J.L., X.Y., S.L. and Y.W.; Methodology, J.P. and P.H.; Project administration, J.P.; Resources, J.P. and J.L.; Software, J.P.; Supervision, J.P. and X.Y.; Validation, J.P.; Visualization, J.P.; Writing—original draft, J.P.; Writing—review & editing, J.P. All authors have read and agreed to the published version of the manuscript.

Funding: This work was supported by the National Natural Science Foundation of China (52071165), the National Natural Science Foundation of China (51771083), the Henan Province Key R & D and Promotion Project (Science and Technology Research) (222102220082), Henan Postdoctoral Research Project (202003076), the Henan Excellent Youth Fund Project (202300410268), the Scientific and technological innovation talents in Colleges and universities in Henan Province (22HASTIT026), special support for high-level talents in Henan Province “Central Plains Talent Program—top young talents in Central Plains” (ZYBJ202126).

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: Not applicable.

Conflicts of Interest: The authors declare no conflict of interest.

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